

SET-D | Quiz-2 | CSE425.10 | NSU | Date: 25.7.26

1. What is the primary task of the Lexical Analysis phase of a compiler?

- A. Building a parse tree from tokens
- B. Grouping characters of the source code into tokens and removing whitespace
- C. Checking that operand types are compatible
- D. Translating intermediate code into machine instructions

2. Which compiler phase takes a stream of tokens and builds a tree structure that represents the grammatical structure of the program?

- A. Lexical Analysis
- B. Parsing
- C. Optimization
- D. Code Generation

3. Verifying that a program's statements are meaningful (e.g., type checking, matching operand types) is the job of which phase?

- A. Parsing
- B. Semantic Analysis
- C. Code Generation
- D. Lexical Analysis

4. Which phase improves the intermediate representation so the program runs faster or uses fewer resources, without changing what it computes?

- A. Lexical Analysis
- B. Semantic Analysis
- C. Optimization
- D. Parsing

5. Which phase is the final stage of compilation, producing target machine instructions from the (optimized) intermediate representation?

- A. Code Generation
- B. Parsing
- C. Lexical Analysis
- D. Semantic Analysis

6. What is the correct order of the five basic compiler phases?

- A. Parsing → Lexical Analysis → Semantic Analysis → Code Generation → Optimization
- B. Lexical Analysis → Parsing → Semantic Analysis → Optimization → Code Generation
- C. Lexical Analysis → Semantic Analysis → Parsing → Optimization → Code Generation
- D. Semantic Analysis → Lexical Analysis → Parsing → Code Generation → Optimization

7. In regular expressions, what does the dot metacharacter (.) match?

- A. Only a literal period character
- B. Any single character
- C. Zero or more characters
- D. Any digit character

8. Which of the following correctly matches a literal period character in a regex pattern?

- A. a.b
- B. a\.
- C. a[.]*b
- D. a^.

9. Which shorthand character class matches any single digit character?

- A. \d
- B. \w
- C. \d
- D. \s

10. What does the pattern [abc] match in a regex?

- A. The exact string "abc"
- B. Any one of the characters a, b, or c
- C. Zero or more occurrences of a, b, c
- D. None of the characters a, b, c

11. What does the pattern [^abc] match?

- A. Any character that is not a, b, or c
- B. Only the characters a, b, or c
- C. The literal text "abc"
- D. Any character immediately followed by "abc"

12. The character range [a-z] in a regex matches:

- A. Only the characters 'a' and 'z'
- B. Any single lowercase letter from a to z
- C. Any character except a and z
- D. Any digit from 0 to 9

13. Which shorthand character class matches any alphanumeric character (letters, digits, or underscore)?

- A. \s
- B. \W
- C. \w
- D. \D

14. In regular expressions, \s matches:

- A. Any whitespace character such as space, tab, or newline
- B. Any non-whitespace character
- C. A literal letter 's'
- D. Any digit character

15. Which pattern correctly matches only the exact standalone word "cat" and not "cats" or "concatenate"?

- A. cat
- B. ^cat\$
- C. .cat.
- D. [cat]

16. What does the pattern [0-9]{3,5} mean?

- A. Exactly 3 digits followed by 5 more digits
- B. Between 3 and 5 repetitions of a digit
- C. At least 5 digits
- D. Exactly 35 digits in a row

17. Which metacharacter matches zero or more repetitions of the preceding element?

- A. +
- B. ?
- C. *
- D. {1,}

18. Which metacharacter makes the preceding character or group optional (matching it zero or one time)?

- A. *
- B. +
- C. ?
- D. {0}

19. Given the pattern (abc|def), which strings would it match?

- A. Only "abcdef"
- B. Either "abc" or "def"
- C. "abc" followed by "def" repeated
- D. Neither "abc" nor "def"

20. In the pattern (a(bc)), what does the inner group (bc) represent?

- A. An independent match unrelated to the outer group
- B. A nested capture group that captures "bc" within the larger match "abc"
- C. An optional character in the pattern
- D. A character exclusion class